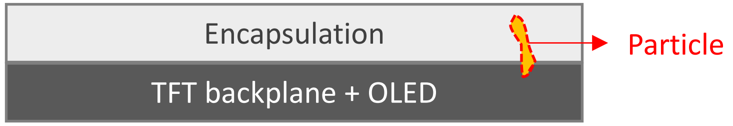
Reduced PECVD Particle Defects via RPSC NF_3 Flow Optimization

Problem

- Persistent particle defects
  - Recurring particle defects were observed
  - The defects caused reliability failures, scrap, and yield loss

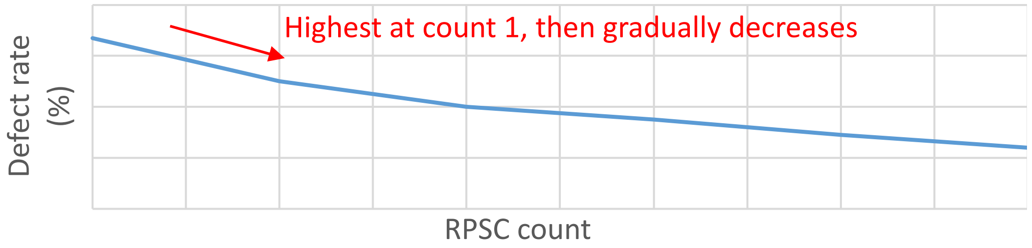
Investigation

- FIB-SEM analysis
  - Identified the PECVD encapsulation layer as the defect source



[Cross-section diagram]

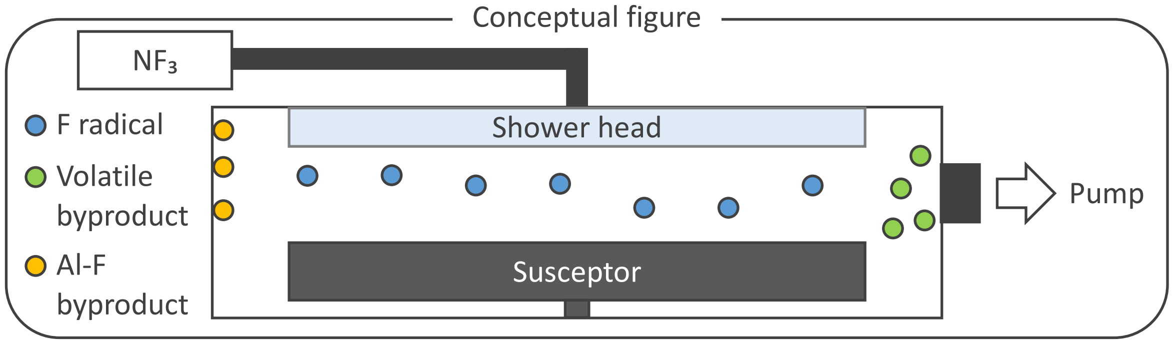
- Defect trend analysis by RPSC count
  - The defect rate was highest at RPSC count 1 and gradually decreased over subsequent counts



  - This trend suggests that **post-clean residual byproducts** were depleted during the early RPSC counts
  - After a fixed number of counts, chamber cleaning is performed, and the count sequence restarts

Hypothesis

- NF_3–chamber material interaction
  - NF_3 species reacted with Al-containing chamber materials during RPSC, forming **Al-F byproducts** that acted as post-clean particle sources



  - Volatile Si-related byproducts (e.g., SiF_4) are pumped out, whereas Al-F byproducts can remain as **post-clean particle sources**

Verification

- Lower NF_3 flow
  - Reduced particle defects during the early RPSC counts
- EDS analysis
  - Showed strong Al and F signals, supporting the proposed chamber-material fluorination mechanism

Action

- NF_3 flow optimization
  - Adjusted the NF_3 flow during RPSC to suppress post-clean particle spikes and reduce the overall defect rate

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